

Heart Failure — Clinical Reference Summary

Classification, Staging, Diagnosis, and Management

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1. Definition and Classification

Heart failure is a clinical syndrome caused by structural or functional cardiac abnormality that results in elevated intracardiac pressures and/or inadequate cardiac output, producing symptoms such as dyspnea, fatigue, and fluid retention.

1.1 Classification by Ejection Fraction

Category	Left Ventricular Ejection Fraction
HFrEF (reduced)	≤40%
HFmrEF (mildly reduced)	41–49%
HFpEF (preserved)	≥50%
HFimpEF (improved)	Previously ≤40%, now >40% with ≥10-point rise

2. Staging (ACC/AHA-style Stages)

Stage	Description
A	At risk for heart failure; no structural disease or symptoms
B	Structural heart disease present; no symptoms
C	Structural disease with current or prior symptoms
D	Advanced/refractory heart failure requiring specialized intervention

3. NYHA Functional Classification

Class	Functional Description
I	No limitation of physical activity
II	Slight limitation; comfortable at rest, ordinary activity causes symptoms
III	Marked limitation; comfortable at rest, less-than-ordinary activity causes symptoms
IV	Symptoms at rest; discomfort with any physical activity

4. Diagnostic Evaluation

- Clinical history and examination for congestion (elevated jugular venous pressure, edema, rales) and low-output signs.
- Natriuretic peptide testing (BNP or NT-proBNP) to support or refute the diagnosis.
- Transthoracic echocardiography for ejection fraction, chamber size, valvular function, and diastolic parameters.
- ECG for rhythm, conduction disease, and evidence of prior infarction.
- Selective use of cardiac MRI, coronary angiography, or biopsy when an underlying etiology (e.g., infiltrative disease) is suspected.

5. Pharmacologic Management of HFrEF

Four foundational medication classes are generally introduced together at low dose and up-titrated as tolerated, rather than added sequentially over many months, based on evidence that early combination therapy yields greater risk reduction:

- Angiotensin receptor–neprilysin inhibitor (preferred), or ACE inhibitor/ARB if ARNI not suitable.
- Evidence-based beta-blocker (e.g., carvedilol, metoprolol succinate, bisoprolol).
- Mineralocorticoid receptor antagonist (e.g., spironolactone, eplerenone), with potassium and renal monitoring.
- SGLT2 inhibitor, shown to reduce heart failure hospitalization and cardiovascular death across a broad range of ejection fractions.

5.1 Additional/Adjunct Therapies

- Loop diuretics for congestion management, titrated to euvolemia rather than a fixed dose.
- Ivabradine in select patients with elevated heart rate despite maximal beta-blockade.
- Hydralazine/isosorbide dinitrate, particularly in patients with persistent symptoms or specific demographic considerations.
- Device therapy: implantable cardioverter-defibrillator for eligible patients with persistently reduced ejection fraction, and cardiac resynchronization therapy for eligible patients with wide QRS and left bundle branch block morphology.

6. Management of HFpEF

Management emphasizes control of comorbid hypertension, atrial fibrillation, obesity, and coronary disease, alongside diuretics for congestion. SGLT2 inhibitors have shown benefit in reducing heart failure hospitalization in this population as well.

7. Advanced Heart Failure (Stage D)

Options considered in advanced, refractory disease include durable mechanical circulatory support (left ventricular assist device), heart transplantation evaluation in eligible candidates, and palliative/supportive care discussions aligned with patient goals when advanced therapies are not appropriate or desired.

8. Monitoring and Follow-up

- Serial assessment of weight, volume status, renal function, and electrolytes during up-titration.
- Reassessment of ejection fraction periodically to reclassify as improved (HFimpEF) if applicable.
- Structured transitional-care and multidisciplinary disease-management programs to reduce readmission.